MINI PROJECT

AIR QUALITY DATA ANALYSIS AND VISUALIZATION USING PYTHON, POWER BI, AND TABLEAU

Framework For Data Visualization Analytics

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ABSTRACT

This mini project, carried out for the subject Fundamentals of Data Visualization and Analytics (FDVA), analyses an Air Quality dataset using Python, Microsoft Power BI and Tableau. The dataset is stored in a file named AirQuality.csv and contains 9,357 hourly records collected by a gas multisensor device in an Italian city between March 2004 and April 2005. It has 15 original columns, including date and time, reference analyser readings for CO, NMHC, benzene, NOx and NO2, five metal-oxide sensor responses, temperature, relative humidity and absolute humidity. Missing sensor readings are stored in the raw file as -200. Python with Pandas was used to read the semicolon-separated file with decimal commas, replace -200 with null values, check for duplicates, create a Datetime column, extract Year, Month and Hour, and save a cleaned file with 9,357 rows and 19 columns. Matplotlib and Seaborn were then used to create six charts covering missing values, monthly and hourly patterns, the relation between temperature and CO, and a correlation heatmap. Dashboards were also developed in Power BI and Tableau, with KPI cards for average CO, NOx, NO2 and benzene, a monthly CO trend, a pollutant comparison and a temperature-versus-CO scatter plot. In this dataset, CO is lowest in the early morning and peaks at about 8 to 9 am and 7 pm, and it is strongly correlated with benzene (0.93) but only weakly with temperature (0.02). The project shows how preprocessing, charts and dashboards turn raw sensor records into clear information about pollution patterns.

1. Data Preprocessing

The dataset (AirQuality.csv) has 9,357 rows and 15 original columns [1], [2]. It was cleaned in Python using Pandas [3], as shown below, and saved as AirQuality_PowerBI_Ready.xlsx with 9,357 rows and 19 columns.

1.1 Loading and inspecting the data

```
import pandas as pd
import numpy as np

df = pd.read_csv("AirQuality.csv", sep=";", decimal=",")
df = df.dropna(axis=1, how="all")   # empty trailing columns
df.shape        # (9357, 15)
df.info()
```

The raw file uses semicolons as separators and commas as decimal marks, so sep=";" and decimal="," were passed to read_csv. Each line of the original file ends with extra semicolons, which Pandas reads as empty columns, so these were dropped.

1.2 Handling missing values marked as -200

```
num_cols = df.columns.drop(["Date", "Time"])
print((df[num_cols] == -200).sum())
df[num_cols] = df[num_cols].replace(-200, np.nan)
print(df.isnull().sum())
```

Missing sensor readings are stored as -200. This is not a real measurement and is not zero; if it were kept, it would pull averages down and appear as a false low value in every chart. All -200 values in the 13 numeric columns were therefore replaced with NaN. No rows were deleted and no values were filled in, so every chart and KPI uses only the readings that are available.

1.3 Checking duplicates and data types

```
print("Duplicate rows:", df.duplicated().sum())
print(df.dtypes)
```

No duplicate rows were found (0). Date and Time are stored as text, and the 13 measurement columns are numeric (float64). Date and Time had to be combined before the data could be used as a time series.

1.4 Creating Datetime, Year, Month and Hour

```
df["Datetime"] = pd.to_datetime(
    df["Date"] + " " + df["Time"].str.replace(".", ":", regex=False),
    format="%d/%m/%Y %H:%M:%S")
df["Year"] = df["Datetime"].dt.year
df["Month"] = df["Datetime"].dt.month
df["Hour"] = df["Datetime"].dt.hour
```

The time values are written as 18.00.00, so the dots were replaced with colons before Date and Time were combined. The format dd/mm/yyyy was given explicitly so that days and months are not swapped. The records run from 10 March 2004 18:00 to 4 April 2005 14:00 with one record per hour, and every Datetime value is unique. Because of this date range, January and

February come only from 2005, March and April come from both years, and May to December come only from 2004. Month-wise averages in this report therefore combine different periods.

1.5 Summary of the cleaned data

Table 1.3 gives the main statistics of the key variables, calculated from the available (non-missing) readings only.

Table 1.3: Summary statistics of key variables

Variable	Count	Mean	Std	Min	Median	Max
CO(GT) (mg/m³)	7,674	2.15	1.45	0.10	1.80	11.90
NOx(GT) (ppb)	7,718	246.90	212.98	2.00	180.00	1,479.00
NO2(GT) (µg/m³)	7,715	113.09	48.37	2.00	109.00	340.00
C6H6(GT) (µg/m³)	8,991	10.08	7.45	0.10	8.20	63.70
T (°C)	8,991	18.32	8.83	-1.90	17.80	44.60
RH (%)	8,991	49.23	17.32	9.20	49.60	88.70
AH	8,991	1.03	0.40	0.18	1.00	2.23

The averages of CO(GT), NOx(GT) and NO2(GT) in Table 1.3 are the same as the values on the KPI cards of the Power BI dashboard (Figure 3.1).

1.6 Exporting the cleaned dataset

```
df.to_excel("AirQuality_PowerBI_Ready.xlsx", index=False)
df.shape        # (9357, 19)
```

Four columns were added (Datetime, Year, Month, Hour). The cleaned dataset with 9,357 rows and 19 columns was saved as an Excel file and used for the visualizations and dashboards.

2. Data Visualization Using Python

Six charts were created from the cleaned data using Pandas [3], Matplotlib [4] and Seaborn [5] (sns.set_theme(style="whitegrid")). Each chart is explained briefly below. Months are calendar months (Jan to Dec) over the whole record, which runs from March 2004 to April 2005. Where pollutants have different units, they are drawn in separate panels.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
df = pd.read_excel("AirQuality_PowerBI_Ready.xlsx")
```

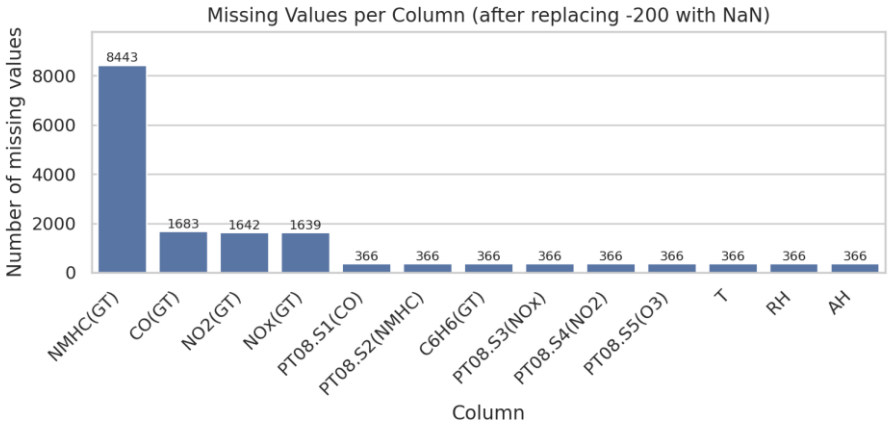


Figure 2.1: Missing values per column (sns.barplot)

Purpose: to show which columns have the most missing readings after -200 was replaced with NaN. Each bar is the number of missing values in one measurement column. **Columns used:** the 13 numeric columns.

Result: NMHC(GT) has the most missing values (8,443), followed by CO(GT) (1,683), NO2(GT) (1,642) and NOx(GT) (1,639). The other nine columns have 366 each.

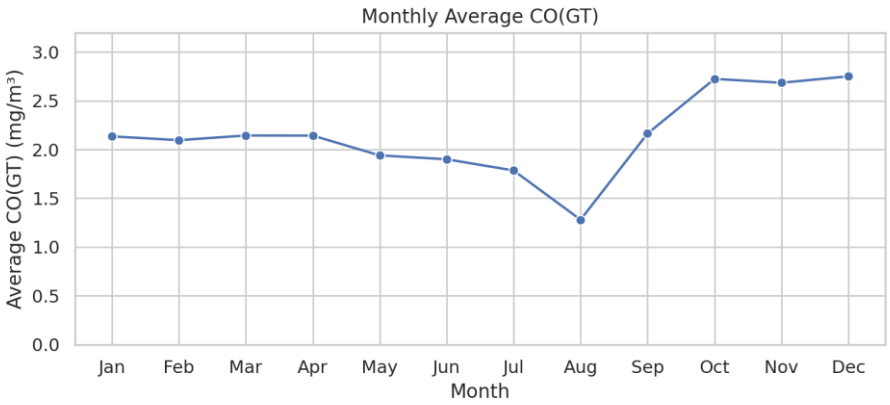


Figure 2.2: Monthly average CO (sns.lineplot)

Purpose: to see how average CO changes through the calendar year. Each point is the mean of all available hourly CO(GT) values in that month.

Columns used: Month, CO(GT).

Result: Average CO is about 2.1 mg/m³ from January to April, drops to its lowest value in August (1.28) and is highest from October to December (2.69 to 2.75). CO is missing for 36% of the hours in August and 47% in October, so these two months rest on fewer readings, and the months come from different years (see Section 1.4).

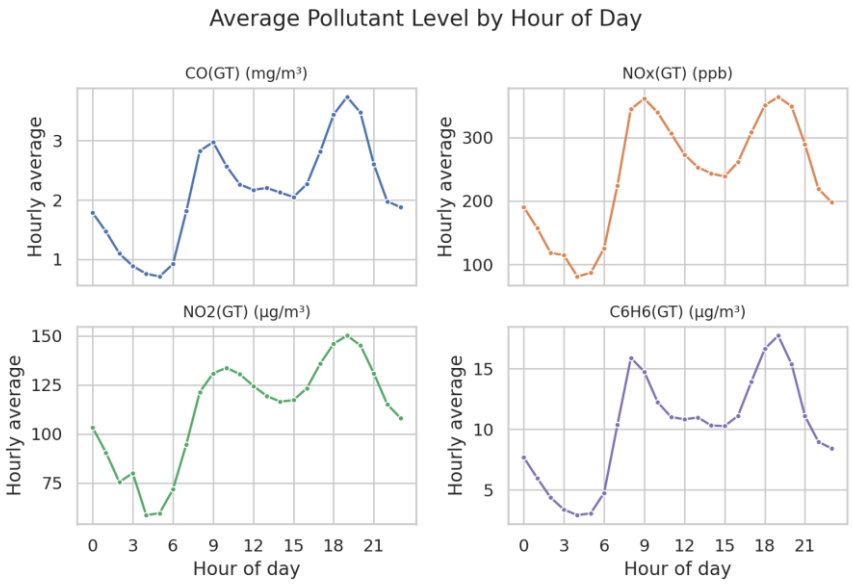


Figure 2.3: Average pollutant level by hour of day (sns.lineplot)

Purpose: to show the daily pattern of each pollutant. Each point is the mean of the available readings at that hour over the whole record.

Columns used: Hour, CO(GT), NOx(GT), NO2(GT), C6H6(GT).

Result: All four pollutants are lowest at about 4 to 5 am. They rise to a morning peak between 8 and 10 am and to an evening peak at about 7 pm. For CO the evening peak (3.73 mg/m³) is higher than the morning peak (2.97 mg/m³). The dataset has no traffic data, so the reason for these peaks cannot be confirmed from it.

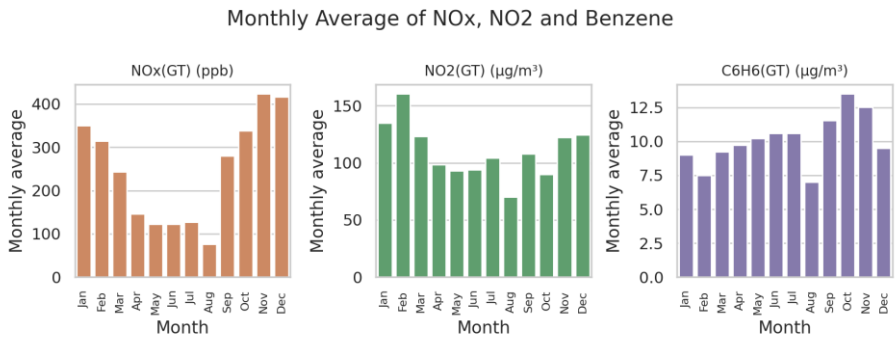


Figure 2.4: Monthly average of NOx, NO2 and benzene (sns.barplot)

Purpose: to compare the monthly pattern of three pollutants, each on its own axis because the units differ.

Columns used: Month, NOx(GT), NO2(GT), C6H6(GT).

Result: NOx is highest in November (424 ppb) and December (417 ppb), NO2 is highest in February (161 µg/m³) and benzene is highest in October (13.5 µg/m³). All three have their lowest monthly average in August.

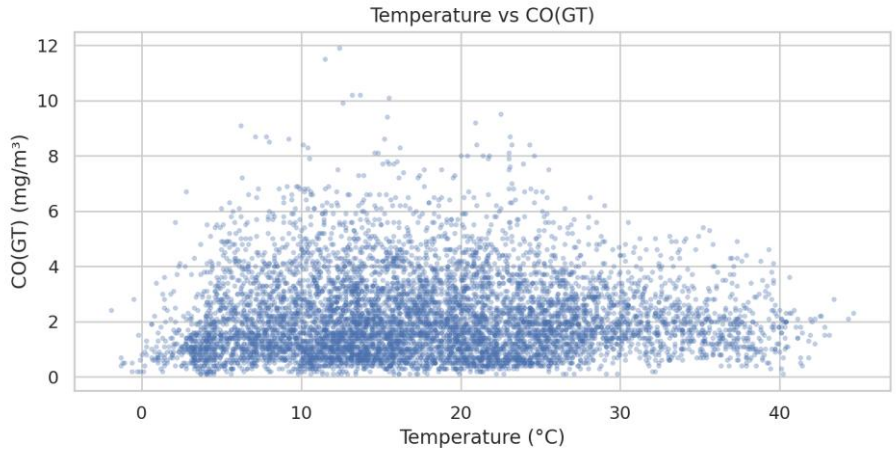


Figure 2.5: Temperature vs CO (sns.scatterplot)

Purpose: to check whether CO changes with temperature. Each point is one hour in which both values are available (7,344 hours).

Columns used: T, CO(GT).

Result: The points form a wide cloud with no clear upward or downward trend, and the correlation is 0.02. CO values above 6 mg/m³ (138 hours) occur between about 3 and 29 °C.

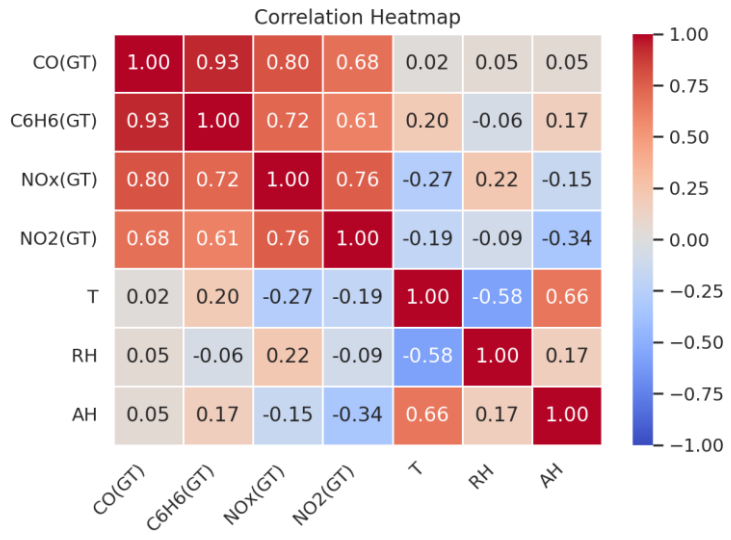


Figure 2.6: Correlation heatmap (sns.heatmap)

Purpose: to show how strongly the pollutants and the weather columns move together. Values close to 1 or -1 indicate a strong relation and values close to 0 indicate almost none.

Columns used: CO(GT), C6H6(GT), NOx(GT), NO2(GT), T, RH, AH.

Result: CO is strongly related to benzene (0.93) and NOx (0.80) and moderately related to NO2 (0.68). Its relation with T (0.02), RH (0.05) and AH (0.05) is very weak. Temperature and absolute humidity are positively related (0.66), and temperature and relative humidity are negatively related (-0.58).

Overall, the pollutants follow a similar daily pattern, and CO moves closely with benzene and NOx but shows almost no linear relation with temperature. A correlation does not show that one variable causes another. The many missing values, especially in NMHC(GT), limit what can be said for some columns.

3. Power BI Dashboard

Power BI [6] was used to build a one-page dashboard from the cleaned air quality data.

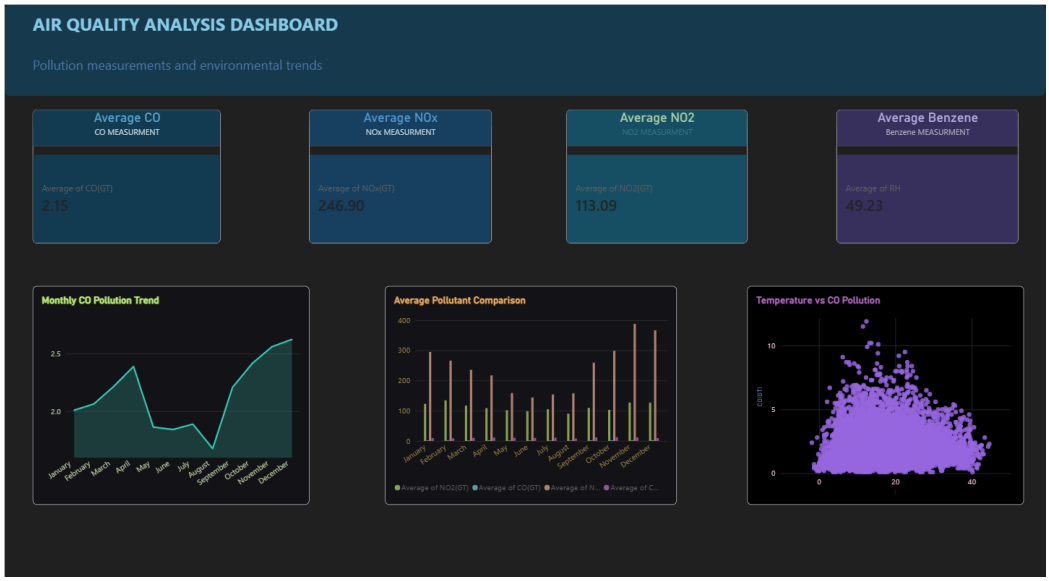


Figure 3.1: Air Quality Analysis Dashboard in Power BI

- **Layout:** a title banner ("Air Quality Analysis Dashboard" with the subtitle "Pollution measurements and environmental trends"), a row of four KPI cards, and a row of three charts.
- **Filters:** no slicers or filters are visible in this screenshot.
- **KPI cards:** Average CO (2.15), Average NOx (246.90), Average NO2 (113.09) and a fourth card titled Average Benzene. Each card shows the average of one field over the whole dataset, ignoring missing values. The cards show no units; in the dataset CO is in mg/m³, NOx in ppb and NO2 in µg/m³.
- **Charts:** Monthly CO Pollution Trend (area line chart of average CO(GT) for January to December); Average Pollutant Comparison (clustered column chart of monthly averages of NO2, CO, NOx and a fourth series whose legend name is cut off in the screenshot); Temperature vs CO Pollution (scatter plot of T against CO(GT)).
- **Insights:** in the line chart, average CO is lowest in August and highest in December. In the column chart, the NOx columns are the tallest in every month and are highest in November, while the CO and fourth-series columns are very small because their values are on a much smaller scale; the chart shows how each pollutant changes across months, not which one is more harmful. The scatter plot shows a broad cloud with no clear trend, which agrees with the correlation of 0.02 in Section 2.

4. Tableau Dashboard

Tableau [7] was used to build the “Air Quality Analysis Dashboard” from the air quality data.



Figure 4.1: Air Quality Analysis Dashboard in Tableau

- **Layout:** a title banner at the top, a filter panel on the left, a row of four KPI cards, and a grid of four charts.
- **Filters:** a date range filter at the top right (1/1/2004 to 12/31/2004) and three drop-down filters on the left for Month, Temperature (T) and Humidity (RH), all set to (All).
- **KPIs:** four cards for Avg. CO, Avg. NOx, Avg. NO2 and Avg. Benzene, each in a different colour with a unit label below the value.
- **Charts:** Monthly CO Pollution Trend (line chart of average CO for January to December); Average Pollutant Comparison (bar chart of CO, NOx, NO2 and Benzene); Temperature vs CO Pollution (scatter plot); and Hourly Pollution Pattern (a heat map of the four pollutants against hour of day, coloured from Low to High).
- **Insights:** this dashboard adds the time-of-day view that the Power BI dashboard does not have, because the heat map shows the relative level of each pollutant for every hour of the day. The filters let the user narrow all charts by date, month, temperature or humidity. As in the Power BI dashboard, the pollutants in the comparison chart have different units in the dataset, so the bar heights should not be read as a measure of health risk.